YEAR 10 SCIENCE (GENERAL)
EXAMINATION REVISION - SEMESTER 2 2016

Natural and Processed Materials

1. The periodic table is useful for chemists as it arranges the elements in a particular way.

   (a) What is a ***group***?

   Column of the periodic table.

   (b) Why are elements put into a particular group?

   They have similar chemical properties - they react the same.

   (c) What are the group numbers of:

   (i) alkali metals? I

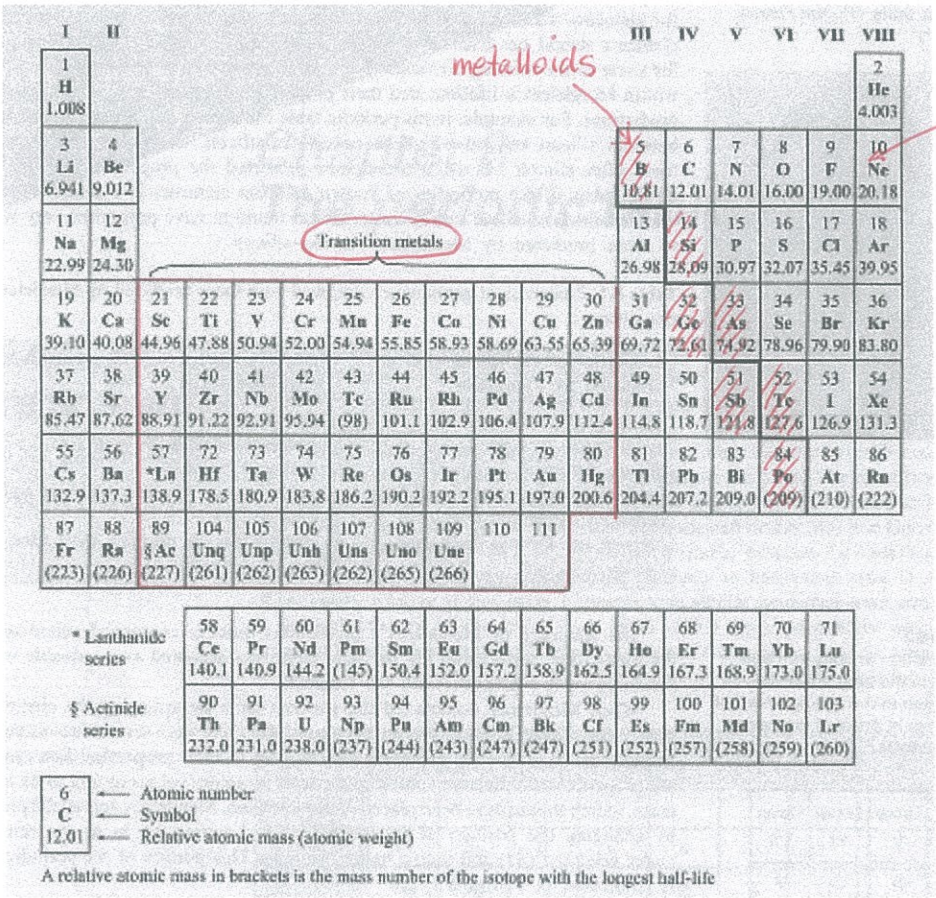
   (ii) inert (noble) gases? VIII

   (iii) halogens? VII

   (d) Where do you find the following elements in the table? Indicate them on the periodic table below, either by labelling or colouring.

   (i) transition elements
   (ii) metalloids / semi-metals

   (e) Where do you find the non-metal elements? Show this on the diagram below.



Non-metals on right-hand-side.

2. Give a definition or description for the following terms.

(a) element — Contains one type of atom.

(b) atomic number — Number of protons.

(c) atomic mass — Number of protons and neutrons

(d) ionic bond — Attraction between oppositely charged ions.

(e) covalent bond — Sharing of two electrons between non-metal atoms.

(f) valence electron — Electron in the outer shell of an atom.

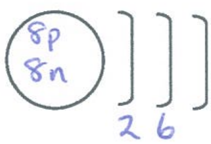
(g) Octet Rule — Atoms donate or share electrons so that they have a completely full outer shell.

3. Complete the following table.

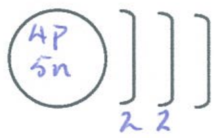
ELEMENT	PROTON NUMBER	NEUTRON NUMBER	ELECTRON NUMBER
$^{32}_{16}S^{2-}$	16	16	18
$^{56}_{26}Fe$	26	30	26
$^{27}_{13}Al^{3+}$	13	14	10

4. Draw clear diagrams to show the electron configurations of the following elements.

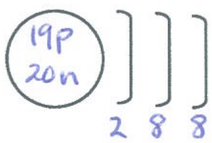
(a) $^{16}_{8}O$



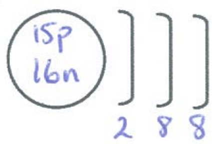
(b) $^{9}_{4}Be$



(c) $(^{39}_{19}K)^{1+}$



(d) $(^{31}_{15}P)^{3-}$



5. $^{16}_{8}O$ can form a 2- ion (O^{2-}).

(a) Were electrons gained or lost to form the O^{2-} ion? gained

(b) How many electrons were gained or lost? 2

(c) In which group is O? VI

(d) What valency do all members of this group possess? 2-

(e) Explain carefully why the $(^{16}_{8}O)^{2-}$ ion is stable, referring to the Octet Rule. (HINT: Some mention of the noble or inert gases would be appropriate!)

- $^{16}_{8}O$ has 6 electrons in its outer shell.
- It has room for 2 more.
- By accepting the 2 electrons, it has a completely full outer shell like the nearest inert gas (Ne) - hence it is stable.

6. Complete the table below by either ***writing a formula or name*** as required. Most are ionic compounds and the rest are covalent molecules. ***For the ionic substances, show the ions used and your working***.

COMPOUND	**FORMULAE**
iron (III) chloride	$FeCl_3$
magnesium iodide	MgI_2
iron (III) nitrate	$Fe(NO_3)_3$
barium carbonate	$BaCO_3$
ammonium sulphide	$(NH_4)_2S$
aluminium hydroxide	$Al(OH)_3$
lead (II) sulphate	$PbSO_4$
potassium acetate	KCH_3COO
zinc phosphate	$Zn_3(PO_4)_2$
copper (ii) sulphate	$CuSO_4$
sulphur trioxide	SO_3
diphosphorus pentoxide	P_2O_5
sulphur hexafluoride	SF_6
nitrogen dioxide	NO_2

7. Balance the following equations.

(a) $MgO_{(s)}$ + 2 $HNO_{3(aq)}$ → $Mg(NO_3)_{2(aq)}$ + $H_2O_{(l)}$

(b) 2 $Al_{(s)}$ + 3 $Cl_{2(g)}$ → 2 $AlCl_{3(s)}$

(c) 3 $Ba(NO_3)_{2(aq)}$ + 2 $K_3PO_{4(aq)}$ → $Ba_3(PO_4)_{2(s)}$ + 6 $KNO_{3(aq)}$

(d) $C_6H_{12}O_{6(s)}$ + 6 $O_{2(g)}$ → 6 $CO_{2(g)}$ + 6 $H_2O_{(g)}$

(e) 2 $NH_{3(g)}$ + 3 $CuO_{(s)}$ → 3 $Cu_{(s)}$ + 3 $H_2O_{(l)}$ + $N_{2(g)}$

8. Write ***balanced equations*** using formulae for the following reactions.

(a) Aluminium metal + oxygen gas → solid aluminium oxide

$4Al + 3O_2 \rightarrow 2Al_2O_3$

(b) Potassium metal + oxygen gas → solid potassium oxide

$4K + O_2 \rightarrow 2K_2O$

(c) Copper (II) sulphate solution + sodium hydroxide solution → solid copper (II) hydroxide + sodium sulphate solution

$CuSO_4 + 2NaOH \rightarrow Cu(OH)_2 + Na_2SO_4$

(d) Solid magnesium hydroxide + hydrochloric acid → magnesium chloride + water

$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$

9. Predict whether a precipitate is produced when the following solutions are mixed together. For those that produce a precipitate, write a ***balanced ionic equation***. Write ***"no reaction"*** if a precipitate is not produced.

(a) potassium sulphate solution + lead (II) nitrate solution → lead sulphate + potassium nitrate

precipitate

$K_2SO_4 + Pb(NO_3)_2 \rightarrow Pb(SO_4)_{(s)} + 2KNO_3$

(b) zinc nitrate solution + sodium phosphate solution → zinc phosphate + sodium nitrate

precipitate

$3Zn(NO_3)_2 + 2Na_3PO_4 \rightarrow Zn_3(PO_4)_2 + 6NaNO_3$

(c) barium nitrate solution + sodium hydroxide solution → barium hydroxide + sodium nitrate

No reaction.

10. Describe ***four*** ways in which the rate of a reaction can be increased.

- Break the substance into smaller pieces.
- Increase the concentration of the reactants or increase the amount.
- Increase the temperature of the reaction vessel.
- Add a catalyst.

11. A conical flask contains 1.0 M HCl at 20 °C. A student adds 2.0 g of Mg metal to it and a reaction occurs, producing H_2 gas. Describe two ways in which the student could increase the rate of production of H_2 gas.

- Use 2.0 M HCl.
- Heat the flask.
- Add 4.0 g of Mg.

Physical Sciences

1. A man drives his car 800 m due East and then turns North, driving 300 m before stopping at a petrol station. It took him 120 s to get there.

(a) What distance he has driven.

$d = 800 + 300 = \underline{1100\ m}$

(b) Calculate his average speed for the journey.

$sp = ?$ $\quad sp = \frac{d}{t}$

$d = 1100 m$ $\quad = \frac{1100}{120}$

$t = 120\ s$ $\quad = \underline{9.17\ m/s}$

2. A balloon filled with hydrogen gas has a velocity of 40 m/s East. How long will it take to go 450 m ?

$sp = 40 m/s$ $\quad sp = \frac{d}{t}$

$d = 450 m$ $\quad \Rightarrow t = \frac{d}{sp}$

$t = ?$ $\quad = \frac{450}{40}$

 $= \underline{11.2\ s}$

3. What is the average speed of a plane if it flies 2700 km in 2.3 hours? Give the answer in km/h.

$sp = ?$

$d = 2700\ km$

$t = 2.3\ hrs$

$$sp = \frac{d}{t}$$

$$= \frac{2700}{2.3}$$

$$= \underline{1174\ km/h}$$

4. A top sprinter reaches 11.2 m/s within 3.15 s of the start of a race. Calculate the average acceleration.

$v = 11.2\ m/s$

$u = 0\ m/s$

$a = ?$

$t = 3.15 s$

$$a = \frac{v-u}{t}$$

$$= \frac{11.2-0}{3.15}$$

$$= \underline{3.56\ m/s^2\ \text{forwards}}$$

5. Calculate the final velocity of a car initially travelling at 10 m/s that accelerates at 1.4 m/s^2 for 10 s.

$v = ?$

$u = 10\ m/s$

$a = 1.4\ m/s^2$

$t = 10 s$

$$v = u + at$$

$$= 10 + (1.4)(10)$$

$$= \underline{24\ m/s\ \text{forwards}}$$

6. Determine the average acceleration of a car that increases its velocity from 7.0 m/s to 34 m/s in 9.0 s.

$v = 34 m/s$

$u = 7 m/s$

$a = ?$

$t = 9.0 s$

$$a = \frac{v-u}{t}$$

$$= \frac{34-7}{9.0}$$

$$= \underline{3.0\ m/s^2\ \text{forwards}}$$

7. A truck moving at 36 m/s is decelerated uniformly at 1.5 m/s^2. What is its final velocity after 5.0 s?

Take forwards as +ve.

$v = ?$

$u = 36 m/s$

$a = -1.5 m/s^2$

"a" is negative - truck decelerates.

$t = 5.0 s$

$v = u + at$

$= 36 + (-1.5)(5)$

$= 28.5$ m/s forwards

8. A car of mass 1530 kg is travelling at 70.0 km/h along Hepburn Avenue when the driver has to brake sharply to avoid a small child who suddenly appeared on a bike crossing the road.

(a) Convert the velocity of the car to m/s.

$v = \frac{70.0}{3.6} = 19.4$ m/s

(b) If it took 3.5 s to stop the car, calculate the average force exerted by the brakes.

$v = 0 ms^{-1}$

$u = 19.4 m/s$

$a = ?$

$t = 3.5 s$

$a = \frac{v - u}{t}$

$= \frac{0 - 19.4}{3.5}$

$= -5.54 m/s^2$

$F = ?$

$m = 1530 kg$

$a = -5.54 m/s^2$

$F = ma$

$= (1530)(-5.54$

$= 8476$ N backwards

(c) The driver of the car was an engineer and had a hard-hat sitting on the back shelf. Describe what happened to the hard-hat in this situation and explain why it took place.

- Hard-hat moved forwards.
- It has inertia and continues forwards while the car is stopping.

9. (a) Explain the difference between ***mass*** and ***weight***. Give an example to help your explanation.

MASS: a measure of the amount of matter in an object.

WEIGHT: force due to gravity on an object.

(b) The acceleration due to gravity on Jupiter is 22.5 m/s^2. Calculate the weight of a 450 kg object on Jupiter.

$F_w = ?$

$m = 450$ kg

$g = 22.5$ m/s^2

$F_w = mg$

$= (450)(22.5)$

$= \underline{10{,}130 \text{ N}}$

10. A 70.0 kg cyclist is applying a force of 150 N forwards but is working against a wind producing a friction force of 80 N opposing the movement.

(a) Determine the resultant force for this situation.

$\Sigma F = 150 - 80$

$= \underline{70 \text{ N forwards}}$

(b) Calculate the acceleration of the cyclist.

$\Sigma F = 70$N

$m = 70.0$ kg

$a = ?$

$\Sigma F = ma$

$\Rightarrow a = \dfrac{\Sigma F}{m}$

$= \dfrac{70}{70.0}$

$= \underline{1.0 \text{ m/s}^2 \text{ forwards}}$

(c) It is possible for the rider to move at constant velocity. In terms of the forces acting, explain how this is possible.

- Yes
- The rider applies 80 N forwards to equal friction.
- $\Sigma F = 0$ and no acceleration occurs.

11. You are sitting on a seat at the moment. Explain why you don't fall through the chair and are supported by it.

- Gravity pulls you down onto the Earth.
- The chair exerts an equal and opposite reaction force upwards
- $\Sigma F = 0$ so you don't move.